

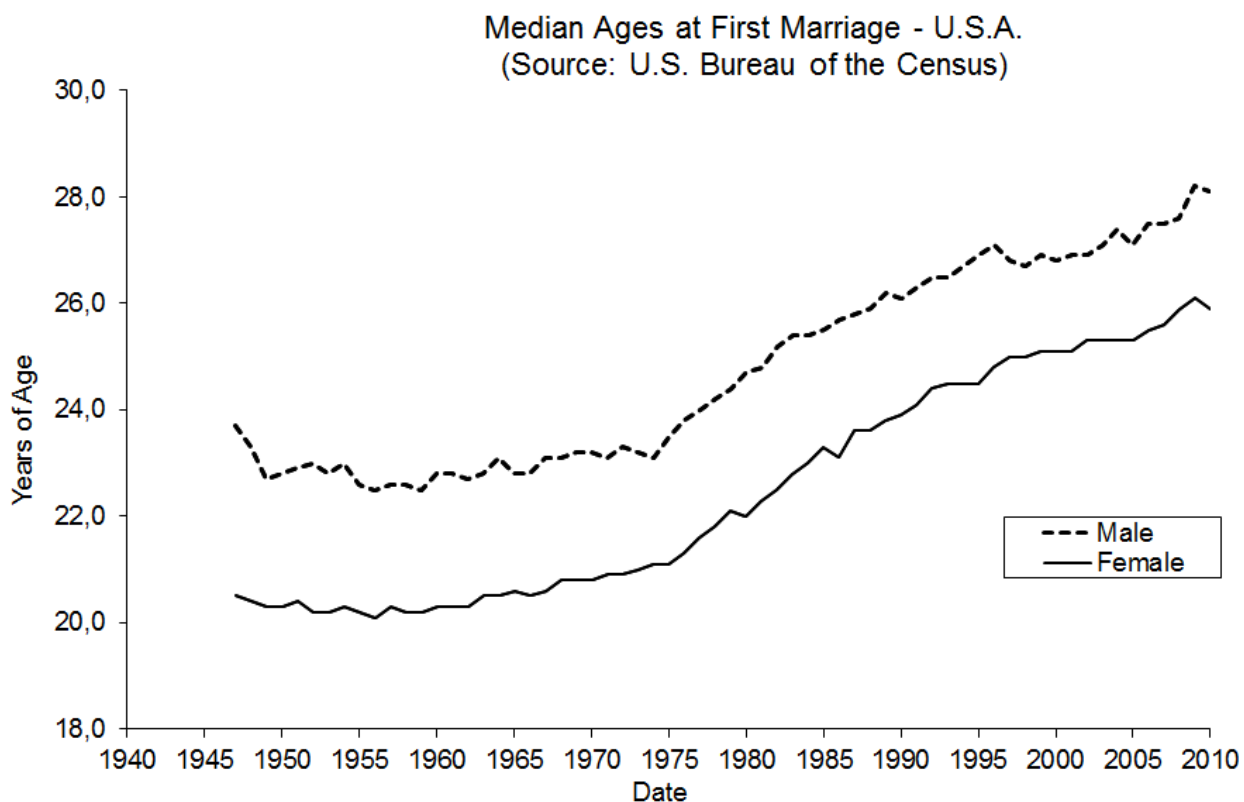
Advanced Econometrics – Exam

September 5, 2019

Exercise 1

Dave Giles at May 30, 2011 published a dataset with estimated median age of Females and Males at first marriage in the USA. In the dataset median age for each year from 1947 to 2010 is stored. The blogger suggested that possible cointegration between the median age for Females (Female) and Males (Male) may exist. To assess that, a number of analyses were performed. Furthermore, it was decided to check whether an Error Correction Mechanism model is appropriate here. Based on the results presented below please answer the following questions:

- Looking at the figure presented below, is there any rationality to consider a possibility that the Female and Male variables are cointegrated?
- Do time series of Female and Male variables satisfy basic assumptions to consider a cointegration between them (please support your answer with appropriate tests results)?
- Is there enough evidence to support a claim that there is cointegration between the variables (please support your answer with appropriate tests results)?
- What is or would be a cointegrating vector for the variables? How to interpret it?
- Does ECM model work? How would you interpret coefficients in the ECM model?



Female median age at first marriage						
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4
0	1.84	0.99	0.25	0.19	0.22	0.10
1	1.95	0.99	0.99	0.33	0.30	0.17
2	1.29	0.99	0.77	0.94	0.80	0.70
3	0.94	0.99	0.79	0.84	0.89	0.63
4	0.49	0.98	0.83	0.63	0.68	0.81
5	0.20	0.97	0.55	0.76	0.86	0.90
						0.11
						0.18
						0.44
						0.49
						0.47
						0.87

First differences for female median age at first marriage						
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4
0	-8.45	0.01	0.92	0.16	0.13	0.07
1	-4.67	0.01	0.66	0.84	0.73	0.68
2	-3.42	0.02	0.72	0.72	0.73	0.56
3	-2.29	0.21	0.79	0.56	0.57	0.69
4	-2.19	0.25	0.54	0.74	0.84	0.87
5	-1.88	0.36	0.79	0.90	0.97	0.99
						0.09
						0.40
						0.46
						0.43
						0.86
						0.99

Male median age at first marriage						
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4
0	1.33	0.99	0.86	0.88	0.28	0.43
1	1.21	0.99	0.33	0.55	0.15	0.21
2	1.32	0.99	0.94	0.96	0.06	0.11
3	0.82	0.99	0.92	0.94	0.89	0.95
4	0.67	0.99	0.86	0.98	0.98	0.98
5	0.56	0.99	0.81	0.96	0.99	0.97
						0.56
						0.32
						0.18
						0.98
						1.00
						0.99

First differences for male median age at first marriage						
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4
0	-7.90	0.01	0.33	0.59	0.11	0.14
1	-6.82	0.01	0.89	0.98	0.04	0.07
2	-4.08	0.01	0.86	0.95	0.91	0.97
3	-3.00	0.04	0.84	0.98	0.99	0.98
4	-2.09	0.29	0.82	0.96	0.99	0.96
5	-2.19	0.25	0.98	0.79	0.91	0.96
						0.23
						0.13
						0.99
						0.99
						0.99
						0.98

Results for the Long-term relationship model

Dependent variable:	

Female	

Male	1.102*** (0.017)
Constant	-4.808*** (0.417)

Observations	64
R2	0.986
Adjusted R2	0.986
Residual Std. Error	0.249 (df = 62)
F Statistic	4,291.233*** (df = 1; 62)
=====	
Note:	*p<0.1; **p<0.05; ***p<0.01

Residuals from Long-term relationship model								
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5	
0	-5.38	0.01	0.57	0.72	0.42	0.59	0.73	
1	-4.28	0.01	0.95	0.97	0.70	0.79	0.88	
2	-2.82	0.07	0.94	0.99	0.95	0.91	0.95	
3	-2.56	0.11	0.96	1.00	1.00	0.99	0.99	
4	-2.11	0.28	0.94	0.93	0.98	0.99	0.99	
5	-2.27	0.22	0.80	0.97	0.99	0.99	0.98	

Results for the ECM model

Dependent variable:	

dFemale	

lresid	-0.270*** (0.070)
dMale	0.307*** (0.082)
Constant	0.068*** (0.017)

Observations	63
R2	0.255
Adjusted R2	0.230
Residual Std. Error	0.125 (df = 60)
F Statistic	10.280*** (df = 2; 60)
=====	
Note:	*p<0.1; **p<0.05; ***p<0.01

Exercise 2

Liuzza et al. (2018) explained authoritarian attitudes with body odour disgust sensitivity¹.

”Authoritarianism has resurfaced as a research topic in political psychology, as it appears relevant to explain current political trends. Authoritarian attitudes have been consistently linked to feelings of disgust, an emotion that is thought to have evolved to protect the organism from contamination. We hypothesised that body odour disgust sensitivity (BODS) might be associated with authoritarianism, as chemo-signalling is a primitive system for regulating interpersonal contact and disease avoidance, which are key features also in authoritarianism.”

”The BODS [is calculated on the basis of variables for] six body odour sources (breath, axillary sweat, feet, faeces, urine and gas), each of which appears in two different contexts: internal source (own smell) and external source (the smell of an unfamiliar person). (p. 3)”

In their third study, the authors showed a positive correlation between BODS scores and support for Donald Trump, with an agenda described as resonating with authoritarian attitudes. Your lecturers decided to estimate binary choice models with the data from the article. The table below presents **logit models’** parameters. Variables used as covariates:

1. *BODS* that stood for body odour disgust sensitivity (continuous; in units; higher values mean more disgust),
2. *GENDER* decoded into dummy variables,
3. *Age* (in years),
4. dummies for education attainment (high school diploma, college without degree, associate degree, bachelor’s degree, master’s degree, professional degree, doctorate degree, and the base level: lower than high school diploma),
5. *SDO* stands for an endorsement of social dominance or inequality, such as social dominance orientation (continuous; in units). The higher the value of *SDO*, the more positive the surveyed person was about social dominance (more likely to ‘step on others to get ahead’),
6. *RWA* – the Right-Wing Authoritarianism (RWA) scale, a well-established assessment measuring authoritarian dispositions (continuous; in units). With higher values of *RWA*, a person agrees more with the right-wing authoritarianism.

The dependent variable *Trump* is binary and takes value of 1 for people who were going to vote for Donald Trump in the presidential elections, and 0 for Hillary Clinton supporters.

¹Marco Tullio Liuzza, Torun Lindholm, Caitlin B. Hawley, Marie Gustafsson Sendén, Ingrid Ekström, Mats J. Olsson, Jonas K. Olofsson, *Body odour disgust sensitivity predicts authoritarian attitudes*, R. Soc. open sci. 5 2018

	<i>Dependent variable:</i>			
	Trump			
	(1)	(2)	(3)	(4)
BODS	0.290* (0.170)	0.336** (0.167)	0.333** (0.164)	0.326** (0.164)
SDO	−0.457*** (0.104)	−0.450*** (0.102)	−0.433*** (0.099)	−0.435*** (0.099)
RWA	−0.590*** (0.114)	−0.570*** (0.112)	−0.577*** (0.112)	−0.578*** (0.111)
GenderM	−0.046 (0.263)	−0.137 (0.256)		
GenderOther	−15.033 (844.241)	−14.630 (858.126)		
Age	0.016 (0.010)	0.015 (0.010)	0.015 (0.010)	
EDU.HighSchool	−1.139 (1.352)			
EDU.CollegWODegree	−0.840 (1.327)			
EDU.AssociateDegree	−0.568 (1.344)			
EDU.BachelorDegree	−1.316 (1.322)			
EDU.MastersDegree	−1.550 (1.366)			
EDU.ProfessionalDegree	−2.351 (1.772)			
EDU.DoctorateDegree	−1.624 (1.696)			
Constant	4.252*** (1.619)	2.946*** (0.933)	2.798*** (0.885)	3.417*** (0.795)
Observations	391	391	391	391
Log Likelihood	-195.666	-199.614	-200.500	-201.683
Akaike Inf. Crit.	419.333	413.227	411.000	411.365

Note:

*p<0.1; **p<0.05; ***p<0.01

- a) Why did your lecturers decided to estimate logit model instead of simple linear regression model?
- b) Select the most appropriate model from the four presented in the table. Explain your choice.
- c) Interpret parameters of the most appropriate model. What is the effect of significant variables?
- d) Can models (1)-(4) illustrate general-to-specific procedure?
- e) Let's assume that instead of *Trump* variable we want to model *candidate*. Our new variable codes for whom a person is going to vote for in the presidential elections. It takes values in the set {Donald Trump, Hillary Clinton, Gary Johnson, Jill Stein and Darrel Castle}. What is the most appropriate kind of models in this case?
- f) Now, let's imagine that we want to build a model for a different variable named *FiscCons*. The variable codes attitudes towards fiscal conservatism. Participants of the survey were asked the question "When it comes to fiscal issues, how would you describe your political attitudes?". Participants reported their responses on a seven-point scale ranging from 1 (very conservative) to 7 (very liberal). What kind of models is the most appropriate for *FiscCons*?

Exercise 3

Kieran O'Connor and Amar Cheema in their paper "Do Evaluations Rise With Experience?" verified the hypothesis that professors gave higher grades the longer they offered the same course. Although the paper is publicly available, the data on courses' grades is confidential. Therefore, your lecturers estimated similar models with a different dataset.

The dataset for your exercise consisted of students' grades from a different university than the one of O'Connor and Cheema. Information on all courses given since Fall 2001 were collected. Some of the courses were offered since Fall 2001, some since Spring 2001, and so on. The courses differ in the length of their offerings. We removed from the database all the courses that were offered fewer than 4 times.

The table below presents results of five panel data models. Column (1) consists of fixed effects model, while columns (2)-(5) comprise random effects models' estimates. The dependent variable is *AvScore* that codes the average score of the given semester course. For a given semester students' results (on the 0-100 scale) were averaged. The set of independent variables consists of *Noffered* that assigns a consecutive number to each consecutive semester that the course is offered, *Sex* that codes the sex of a lecturer (0 for males, 1 for females), *Semester* – a dummy variable indicating a semester (0 Fall, 1 Spring), and *Year* included to account for the passage of time.

The R output below comes from the model with the fixed effects estimator and the Hausman test for models (1) and (2). The model had the form

$$AvScore_{it} = \beta_1 Noffered_{it} + \beta_2 Sex_{it} + u_i + \varepsilon_{it} \quad (1)$$

- a) Is the panel data in the fixed effects model balanced?
- b) Why was parameter for *Sex* not estimated?
- c) Which model among (1) and (2) should be chosen?
- d) Let's assume that we believe that the unobserved time-invariant individual effect is attributed to a positive attitude towards other people. We suppose that lecturers with a more positive attitude mark their students with higher marks. Should we in this case use the fixed or random effects estimator?
- e) Interpret the Hausman test's results.
- f) What can go wrong if we decided to use simple regression for this data?
- g) Some argue that when it is green outside and the sky is blue people are in better mood. Verify the hypothesis that professors gave higher grades in Spring semesters.
- h) Does student quality increase each year?
- i) O'Connor and Cheema formulated a hypothesis that courses that award higher grades are more likely to be offered again. Results from column (5) were obtained for the random effects model estimated with courses offered at least 10 times. Comment on model (5) results.

Please don't be discouraged by the fact that dr Chlebus and dr Woźniak gave this course for the first time :-)

	<i>Dependent variable:</i>				
	AvScore				
	(1)	(2)	(3)	(4)	(5)
Noffered	0.499*** (0.002)	0.499*** (0.002)	0.499*** (0.002)	0.504*** (0.052)	0.476*** (0.091)
Sex		−0.648 (0.620)	−0.648 (0.621)	−0.651 (0.623)	−0.313 (0.983)
Semester			0.088*** (0.021)	0.088*** (0.021)	0.151*** (0.030)
Year				−0.010 (0.104)	0.048 (0.183)
Constant		83.424*** (0.445)	83.381*** (0.446)	104.026 (208.854)	−14.468 (366.431)
Observations	3,131	3,131	3,131	3,131	1,504
R ²	0.967	0.967	0.967	0.967	0.967
Adjusted R ²	0.965	0.967	0.967	0.967	0.967
F Statistic	85,625.060***	45,387.610***	30,427.620***	22,813.790***	10,845.400***

Note:

*p<0.1; **p<0.05; ***p<0.01

Oneway (individual) effect Within Model

Call:

```
plm(formula = AvScore ~ as.numeric(Noffered) + Sex, data = courses,
     model = "within", index = c("Id", "Noffered"))
```

Unbalanced Panel: n = 200, T = 4-30, N = 3131

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-1.2738347	-0.4780633	-0.0096102	0.4804250	1.3106235

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
Noffered	0.4992405	0.0017061	292.62	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

F-statistic: 85625.1 on 1 and 2930 DF, p-value: < 2.22e-16

Hausman Test

data: AvScore ~ as.numeric(Noffered) + Sex

chisq = 0.18577, df = 1, p-value = 0.6665

alternative hypothesis: one model is inconsistent

